Academic Collaboration of Excellence in Cybersecurity

ACE
CYBER R&D PROGRAM

Conversational AI Interface for Analyst-Driven Threat Investigation

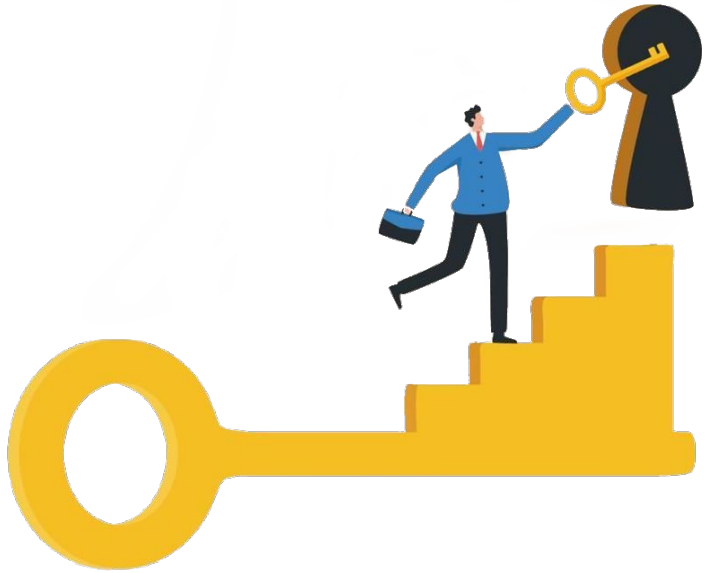
Presented by:
Koi McFarland
Charleston Southern University
11/12/2025

Faculty Advisor: Professor Henderson, CSU
Ace Mentor: Professor Littleton, NIWC

"Collaborative R&D Partnerships for Cybersecurity Excellence"

Executive Summary & Key Accomplishments

- **Key Accomplishment:** Successfully developed and evaluated a Conversational AI Interface (CAI) that translates natural-language threat-hunting queries into valid Elasticsearch JSON DSL, achieving high detection accuracy across multiple threat categories using a fully synthetic, ECS-compliant dataset.



Executive Summary & Key Accomplishments

- **Impact:** This reduces analyst workload and accelerates SOC investigation workflows by eliminating the need for manual query construction.
- **Core Finding:** The CAI achieves high F1-scores for failed logins, foreign-IP detection and lateral movements while demonstrating adept linguistic intelligence. It also enforces safety guardrails which proves the viability of natural-language interfaces for SOC query operations.



The Problem & Its Impact

- **Core Challenge:** SOC analysts must manually construct complex Elasticsearch JSON DSL queries to investigate threats, requiring deep knowledge of field mappings and Boolean logic, which slows response times.
- **Negative Impact:** Analysts lose valuable time crafting and troubleshooting DSL queries, which creates operational bottlenecks and increasing training overhead as SOC teams scale and must teach this specialized syntax.



The Problem & Its Impact

Quotes on Manual SIEM Efforts:

- “This process, which is both time-consuming and prone to human error, requires significant expertise and experience, resulting in the inefficient allocation of human resources and an increased workload for analysts” (Shukla et al., 2025).
- “Writing correct KQL requires specialized expertise, and this dependency creates a bottleneck as security teams scale” (Muzammil et al., 2025).
- “We are using ElasticSearch for storing data, but the GUI/KQL/DSL languages are a real pain” - StackOverflow User

Project Goals & Objectives

- **Primary Goal:** To develop and test a Conversational AI Interface that allows SOC analysts to query Elasticsearch using natural language and evaluate its accuracy against a fully labeled synthetic dataset.



Project Goals & Objectives

- **Key Objectives:**
  1. Build a functional CAI prototype integrated with Elasticsearch and powered by a local LLM via Ollama.
  2. Generate a controlled, fully labeled synthetic dataset to evaluate CAI performance (PI-Key).
  3. Execute a test plan to measure F1-score, precision, recall, and linguistic intelligence.
  4. Validate CAI safety (no unsafe queries, correct handling of ambiguous prompts).
  5. Document limitations and identify required improvements for multi-step, multi-field threat behaviors.

My Research Approach

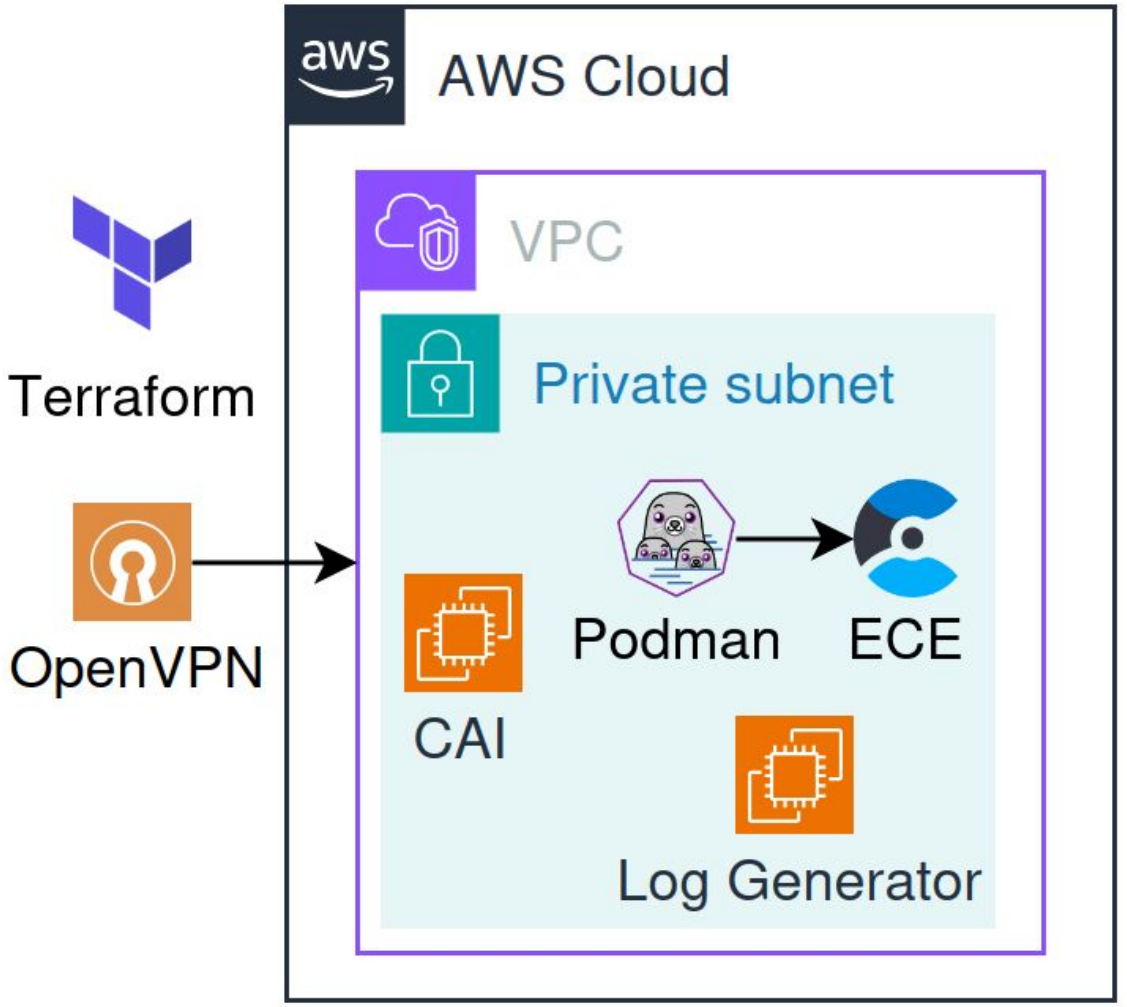
- **Step 1: Build the Environment**

  Deploy AWS-based Elastic Cloud Enterprise architecture using Terraform and Podman.

  Setup the VPC and its subnets.

  Create an EC2 instance for both the log generator and the CAI.

  Setup the VPN and SSH for easy and secure access to the EC2 instances.



My Research Approach

- **Step 2: Create the Generator**

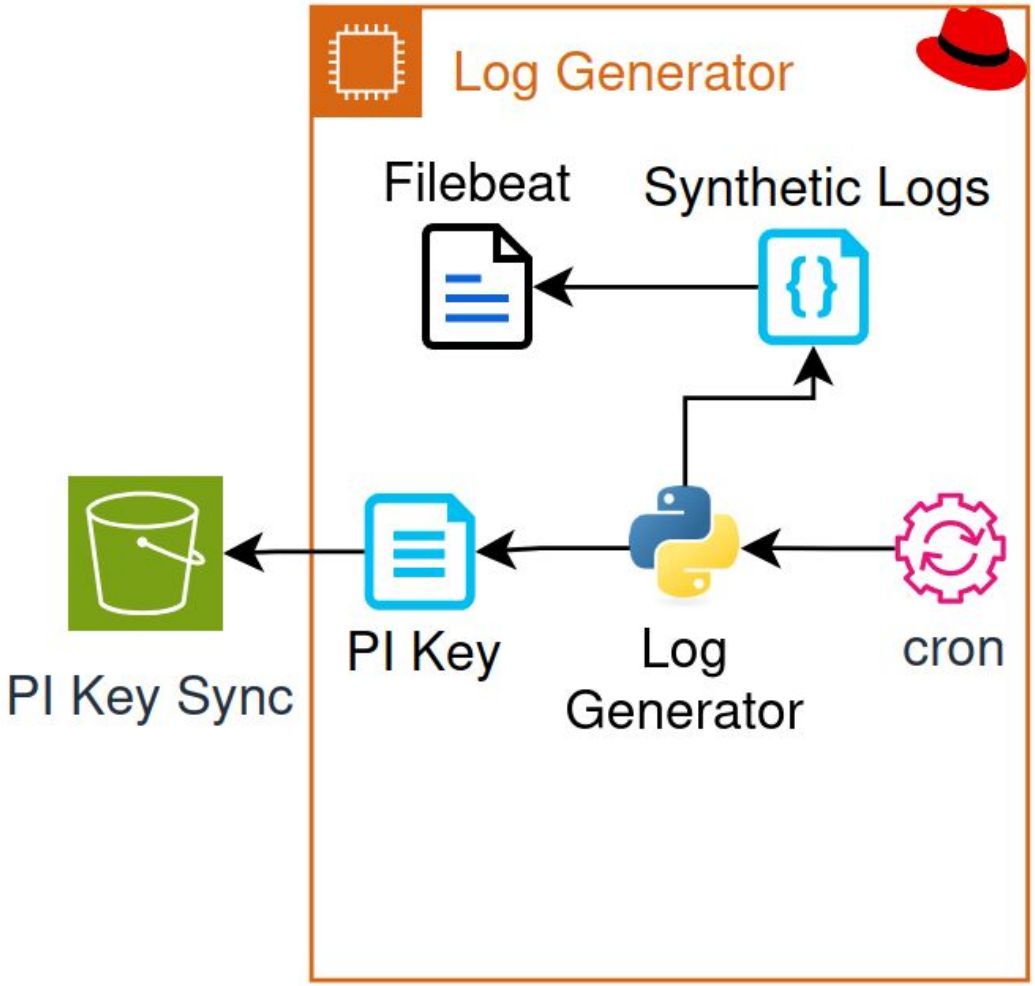
  Make a python script that generates synthetic but realistic logs that are ECS-Compliant.

  Use Filebeat to ingest the logs.

  Start cron to run the script once every 5 minutes.

  Connect each log to a label in a separate PI key file via UUIDs.

  Sync the PI key file to an S3 bucket.



My Research Approach

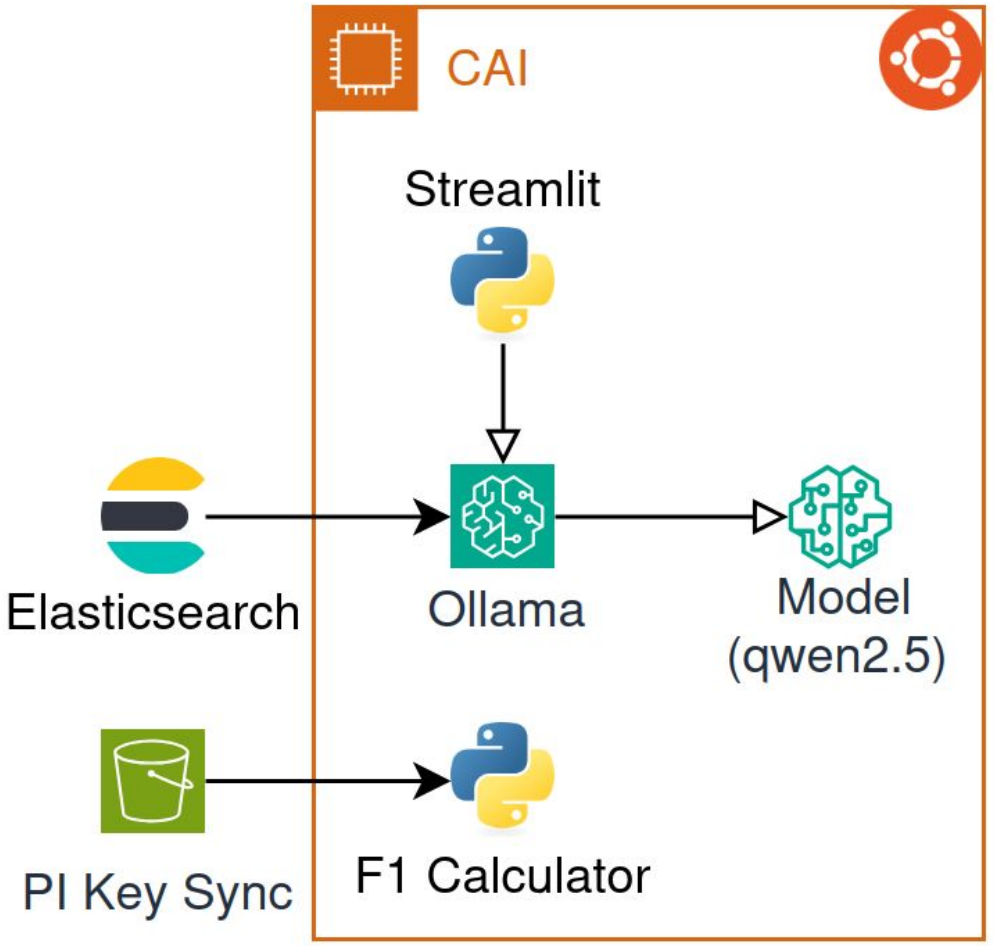
- **Step 3: Create the CAI**

  Configure an EC2 instance with a decent GPU equipped (g5.xlarge).

  Setup Ollama and a model and give it access to Elasticsearch.

  Create an intuitive interface to query the model with python streamlit.

  Create a python script to calculate F1 scores using the PI Key.



My Research Approach

- **Step 4: Run the Test Cases**

  Test the performance of the model/environment.

  Test the model's ability to detect failed logins, lateral movements, and foreign IPs.

  Test the model's lingual robustness.

  Test the model's error handling.



My Research Approach

- **Step 5: Data Analysis**

  Compare CAI responses to PI-Key labels and calculate F1 scores.

  Identify categories where CAI excels vs. fails.

  Validate safety and robustness.

  Record findings in a Research Notebook.



Key Results & Findings

- Now that we understand how the model operates, how did it perform?
- Across the evaluated threat-hunting tasks, the Conversational AI Interface achieved strong overall accuracy, with F1-scores between 0.94 and 0.98.
- The system consistently interpreted natural-language queries correctly and generated valid Elasticsearch DSL that retrieved the majority of ground-truth threats across foreign IP activity, failed-login detection, and lateral-movement patterns.
- Next, we'll examine each test case individually.

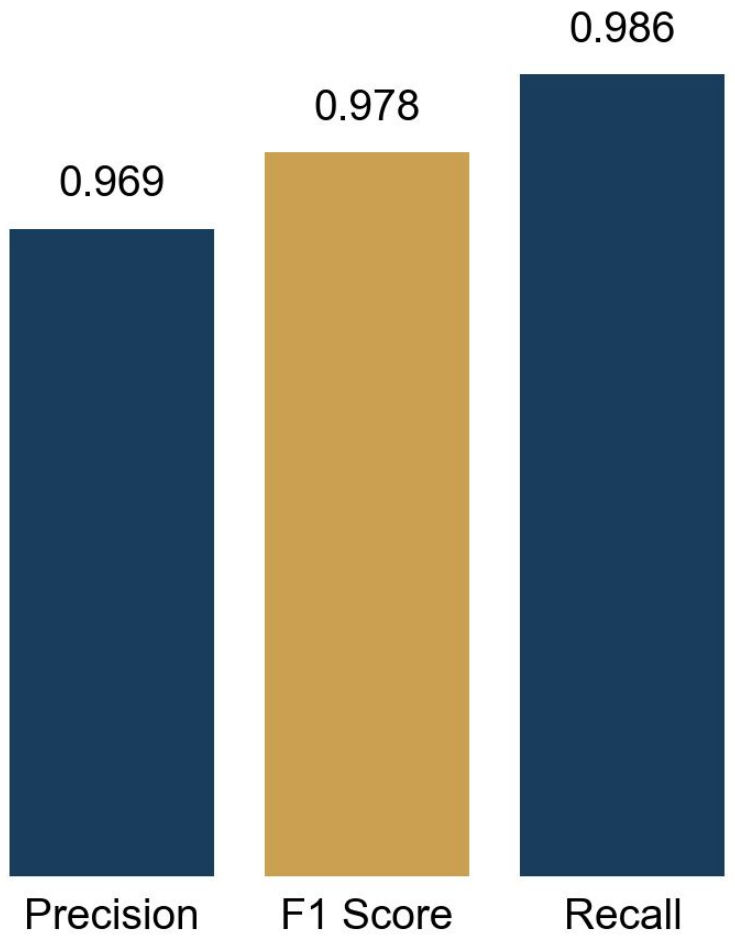
Key Results & Findings

- Here is the Precision, Recall, and F1 Scoring of the model when tasked with finding events that have IPs coming for foreign nations.
- The model uses Geoip data within the Elasticsearch logs to determine which logs are foreign.
- There were:
  - 347 True Positives
  - 11 False Positive
  - 5 False Negatives



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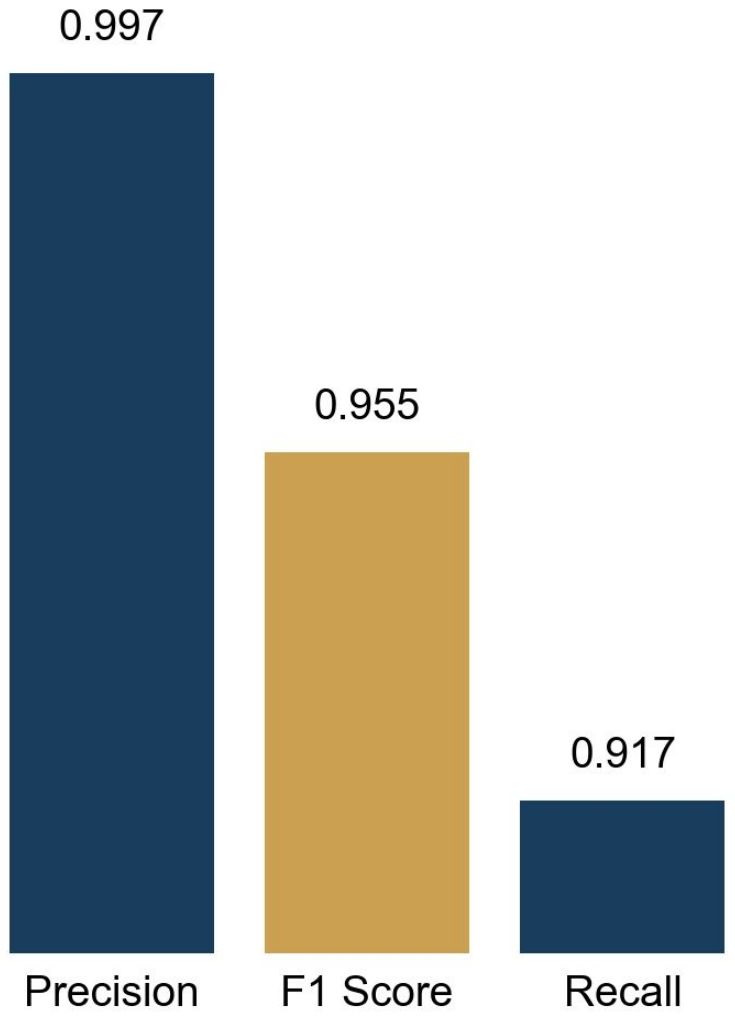
Key Results & Findings

- Here is the Precision, Recall, and F1 Scoring of the model when tasked with finding threatening failed logins events.
- The database includes both benign and adversarial failed logins, so the model was tasked with determining the difference.
- There were:
  - 364 True Positives
  - 1 False Positive
  - 33 False Negatives



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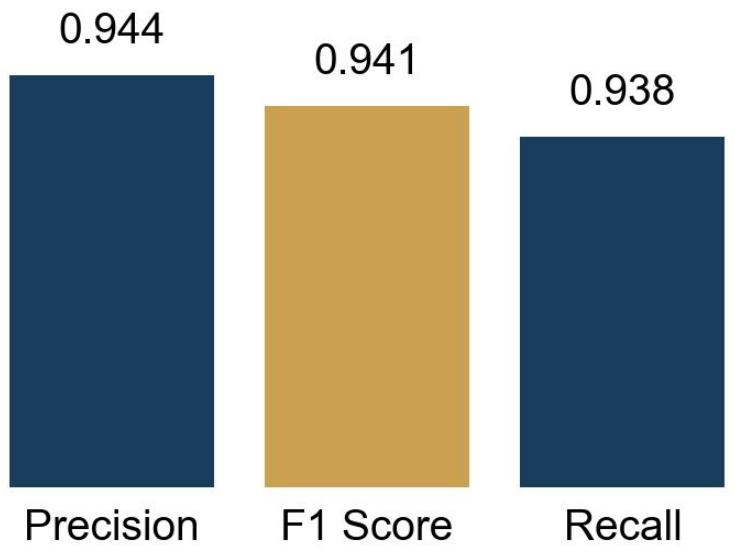
Key Results & Findings

- Here is the Precision, Recall, and F1 Scoring of the model when tasked with finding lateral movements.
- There were:
  - 135 True Positives
  - 8 False Positive
  - 9 False Negatives

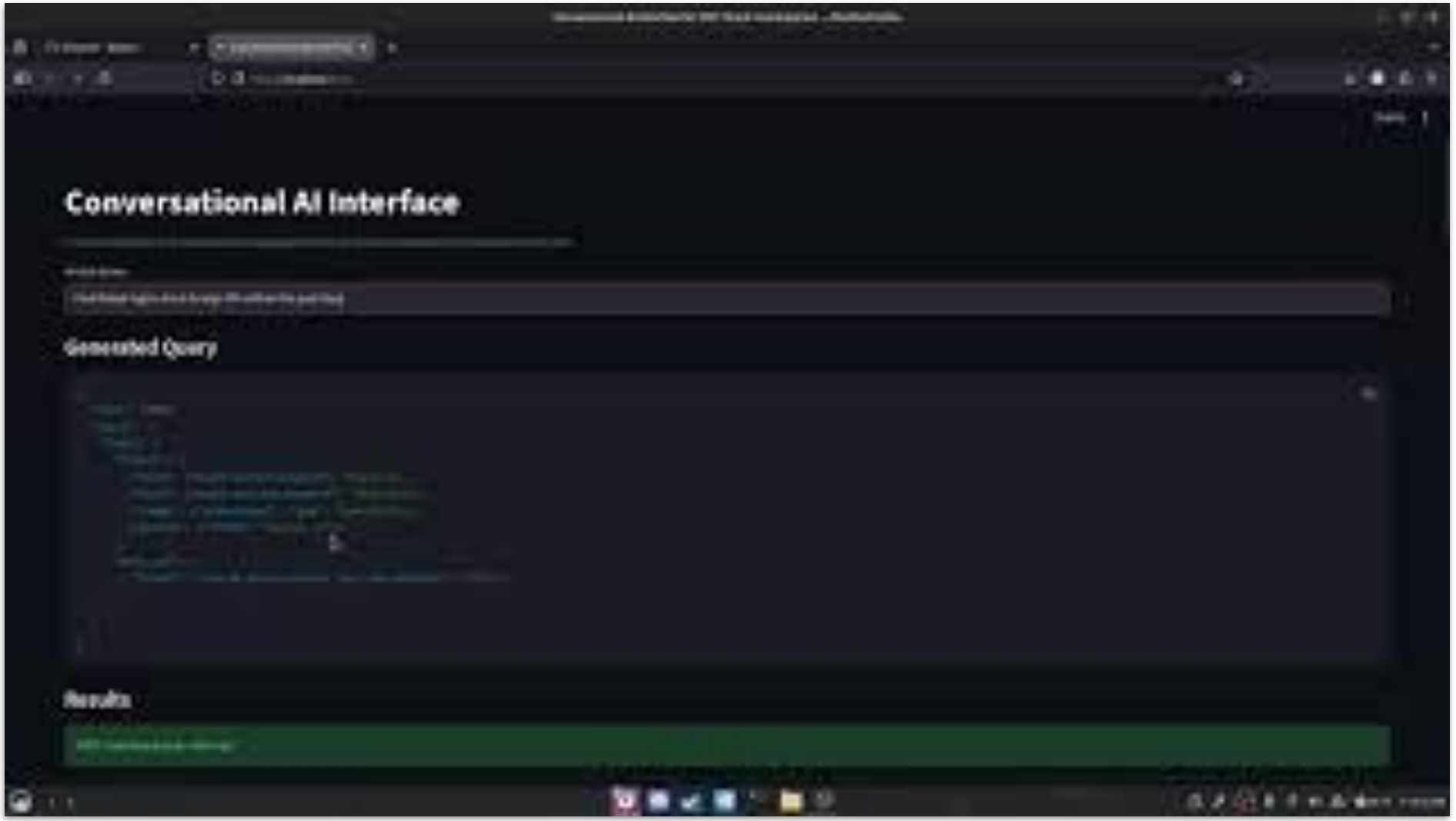


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Demonstration



Conversational AI Interface

Generated Query

Results

Conclusion & Future Work

- **Impact Summary**
  - **Accelerates Investigations:** Cuts query formulation time from minutes to seconds, increasing total alerts an analyst can process per shift.
  - **Expands Analyst Capability:** Allows junior analysts to execute expert-level queries without mastering complex DSL syntax.
  - **Enhances Mission Readiness:** Supports DoW objectives for AI-augmented cyber operations by enabling faster triage, improved detection coverage, and more efficient use of analyst labor-hours.
  - **Demonstrates Viability:** Confirms that natural-language to DSL translation is a practical and safe operational capability for SOC workflows.

Conclusion & Future Work

- **Conclusion**
  This research shows that a constrained LLM can reliably convert natural-language questions into safe, accurate Elasticsearch DSL, reducing time and technical barriers analysts face during investigations.
- **Future Work**
  - Increase Dataset Realism
  - Add Self-Validation Layer
- **Recommendation**

  The DoW should consider investing in the development of a fully operational CAI to evaluate its impact on analyst workload, detection speed, and mission readiness.

Thank You

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Questions?

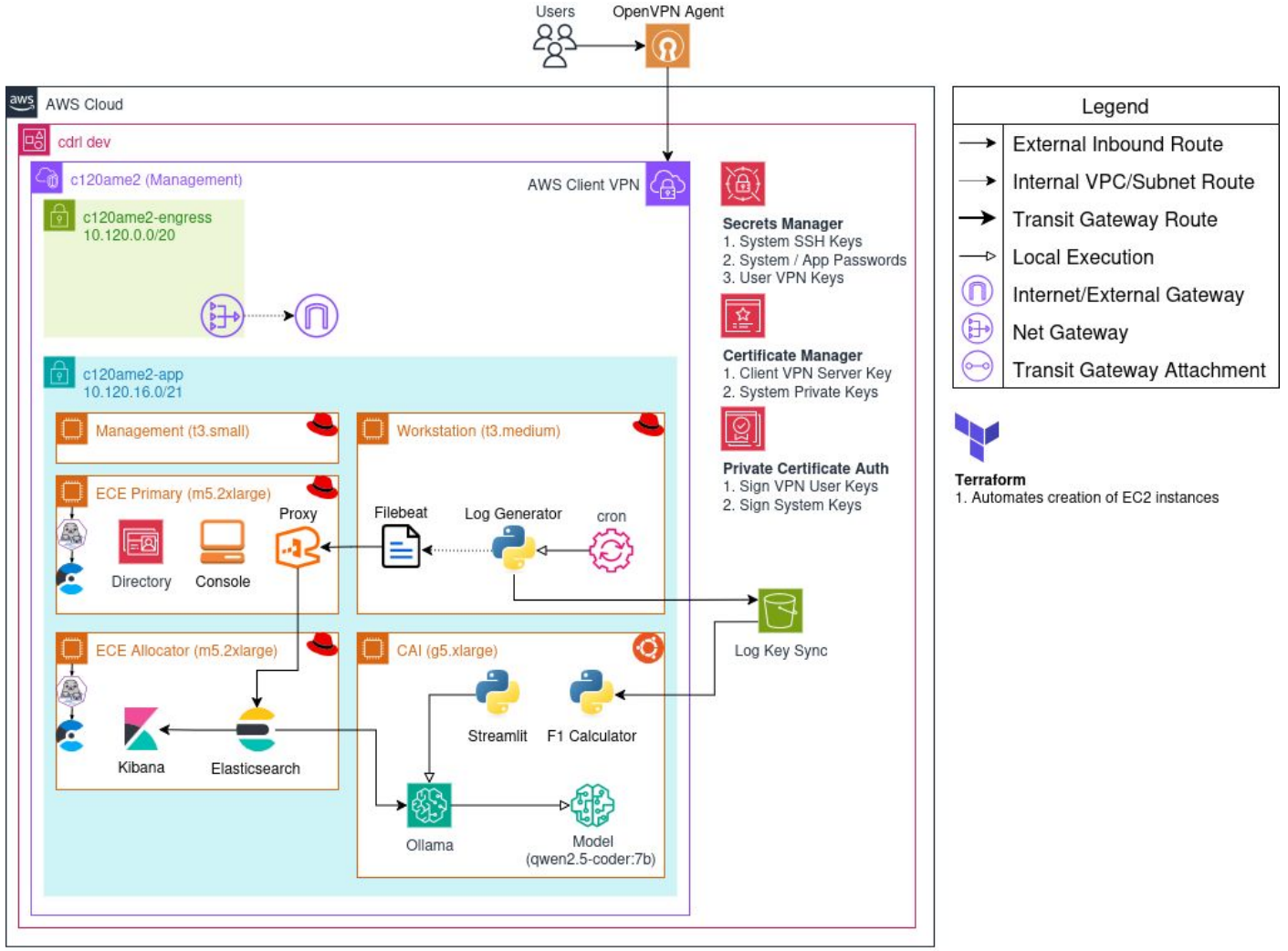


CHARLESTON SOUTHERN UNIVERSITY
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Extra Slides

- Diagram of Testbed Design
- Synthetic Dataset Realism
- Why Synthetic Data?
- CAI vs Traditional SOC Tooling
- Safety Guardrails
- Ollama Model Details
- Current Limitations
- SOC Integration
- Why Local LLM?
- Alignment with DoW goals
- Why a Custom Log Generator?

Diagram of Testbed



Synthetic Dataset Realism & ECS Compliance

- **Standardized fields & hierarchy**
  (event, host, user, source, destination, agent)
- **Normalized timestamps**
  @timestamp in ISO-8601 format
- **Clear entity roles**
  Separation of source, destination, host, and user
- **Structured event semantics**
  event.category, event.action, event.dataset
- **Elastic-compatible enrichment**
  GeoIP fields aligned with ECS
- **Declared ECS version**
  Ensures schema consistency
- **Safe research extensions**
  UUID labeling added without breaking ECS



Why Synthetic Data Was Used

- Synthetic data enables controlled threat insertion, repeatable experimentation, and precise precision/recall measurement without classification constraints or access limitations.
- It also allows full control over threat volume and distribution, including a deliberately elevated ~10% adversarial rate for evaluation.

CAI vs Traditional SOC Tooling

- Traditional SIEM/SOAR workflows require analysts to craft complex query syntax.
- The CAI provides a natural-language interface that translates intent into valid DSL, which augments or adds to existing SOC tooling rather than replacing it.

Safety Guardrails & Query Enforcement

- The CAI applies strict guardrails, permitting only approved ECS fields, constrained operators, and bounded time windows.
- Ambiguous, unsafe, or out-of-policy requests are intercepted before reaching Elasticsearch.

Current Limitations

- The system has not yet been validated in a live SOC environment.
- Model output is not 100% consistent and can rarely fail on straightforward instructions.
- The model cannot self-correct errors and has no idea if the DSL query was successful or not.
- The CAI only supports a limited set of detection categories.

SOC Integration Path

- The CAI integrates alongside existing SOC tooling with no changes to ingestion pipelines.
- Analysts access it through a separate interface, and all queries execute within current SIEM permissions.

Local LLM Selection Rationale

- No data leaves the environment
- Avoids external API dependency
- Ollama specifically is free, making it easier to stick within the project’s budget.

Alignment with DoW AI & Cyber Objectives

- This project aligns with DoW goals for human-in-the-loop AI, analyst augmentation, and responsible AI adoption, improving cyber situational awareness without automating critical decisions.

Why a Custom Log Generator?

- A custom generator was needed to attach UUID-based ground truth to every log, control behavior timing and noise. It also allowed for complete control over everything.

Ollama Model Details

- **Model Architecture:** Qwen2.5-Coder-7B (7-billion parameter LLM)
- **Deployment:** Local inference via Ollama on NVIDIA GPU. (Specifically a g5.xlarge)
- **Model Size:**
  ~4.2 GB (Q4_K_M quantized model used by Ollama)
- **Context Window:**
  32,768 tokens (32K)